

My interpretation of the disaster theory for asset owners is that it reemphasizes the importance of downside outcomes, as I stressed in chapter 2. There, I stressed that mean-variance utility does not cut it in capturing how investors respond to bad events. The disaster story says that those bad events can indeed be very bad, and they are systematically very bad for the entire economy. We need to think about how investors behave in these bad outcomes—a joint statement about the utility function and the data-generating process, which must capture disaster events. Investors who are relatively the best off in severe economic crashes (all investors will be affected adversely during disasters; it is a question of how much) are in the best position to hold large amounts of equities. The overall message is that investors need to think about how bad these events can be, how they can respond to meeting their liabilities, and how they react to these risks when these bad events arrive.

Disasters suggest that another explanation for the high equity premium is that it is a result of markets that have survived *ex post*. In the early 1900s, you could have invested in China or Russia, and you would have lost everything. You could also have invested in Hungary, Czechoslovakia (which were together part of the Austro-Hungarian Empire), Poland, and Greece, but you would not have been able to trade equities again in those countries until the late twentieth century. Jorion and Goetzmann (1999) argue that the realized equity premium is high because we compute the equity return on countries that have survived. Had we computed the returns including all the countries whose markets disappeared, then the premiums would be much lower. Thus, survivorship bias makes the equity premium too high. The true equity premium, which is the one that is relevant for the future, is lower.

I believe this argument only increases the mystery of the equity premium. The equity premium is measured relative to bonds or bills, and sovereign bonds in many countries fared worse than equities. If you invested in German equities in 1900, there were long periods during which markets were closed (like the credit crisis of the early 1930s, World War II, and the occupation by the Allies after World War II), but equity markets survived and eventually German equity holders did well. Holders of German bonds got nothing. First they were ravaged by hyperinflation in the 1920s, and then there was a default on Reichsmark claims when the Deutschmark was introduced in 1948. Of course, in some countries, including Russia and China, both equity holders and debt holders were wiped out. But, in many countries with structural breaks, equity holders eked out a (sometimes modest) long-term return while bond holders were decimated. If the equity premium is measured relative to bonds, this survivorship bias of markets exacerbates the equity premium puzzle. The historical superiority of equities relative to bonds in disasters does carry some implications for factor investing, which I discuss in chapter 14.